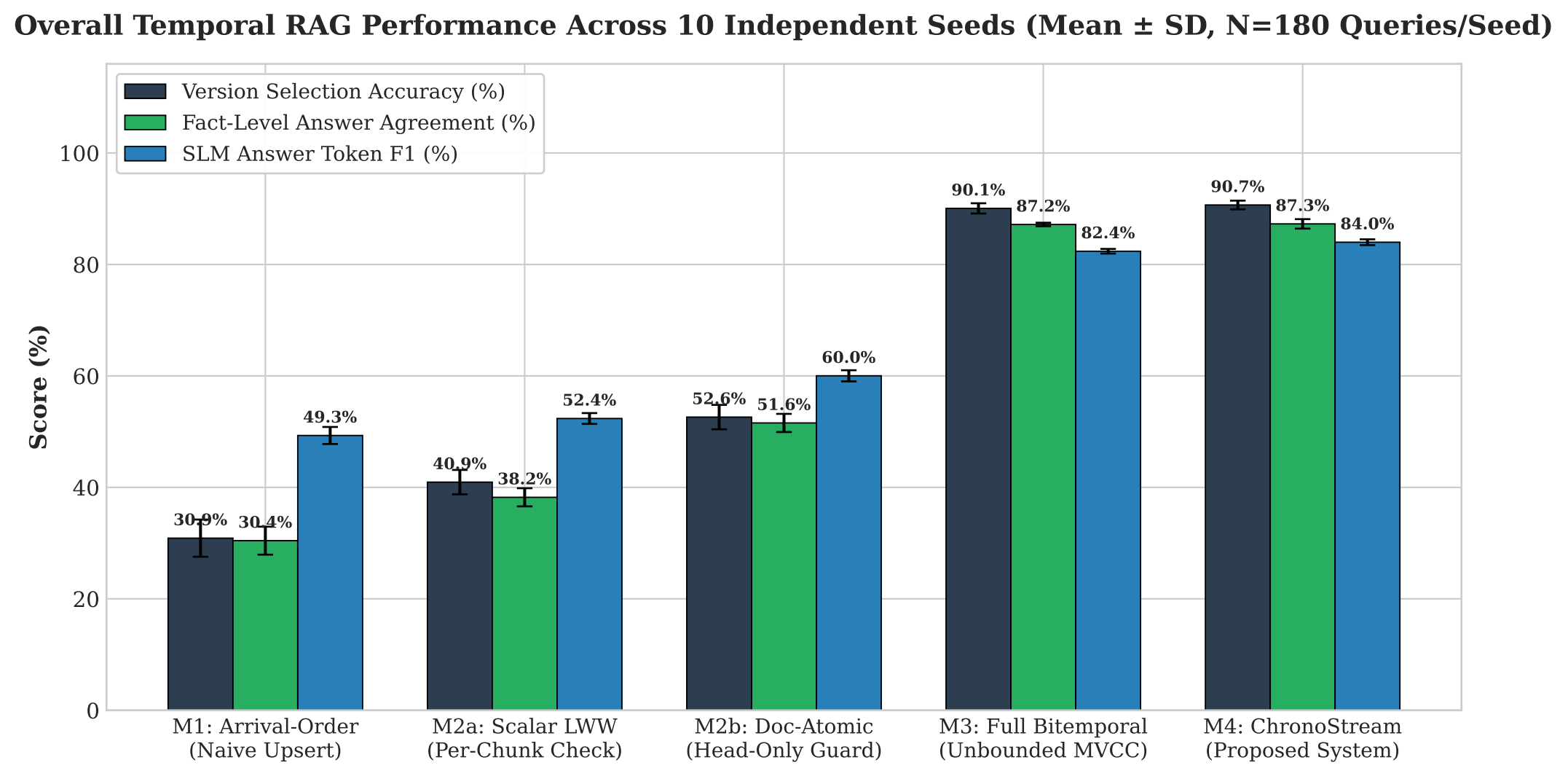
Overall Temporal RAG Performance Across 10 Independent Seeds (Mean ± SD, N=180 Queries/Seed)
Version Selection Accuracy (%)
Fact-Level Answer Agreement (%)
SLM Answer Token F1 (%)
Score (%)
0
20
40
60
80
100
30.9%
30.4%
49.3%
40.9%
38.2%
52.4%
52.6%
51.6%
60.0%
90.1%
87.2%
82.4%
90.7%
87.3%
84.0%
M1: Arrival-Order
(Naive Upsert)
M2a: Scalar LWW
(Per-Chunk Check)
M2b: Doc-Atomic
(Head-Only Guard)
M3: Full Bitemporal
(Unbounded MVCC)
M4: ChronoStream
(Proposed System)